

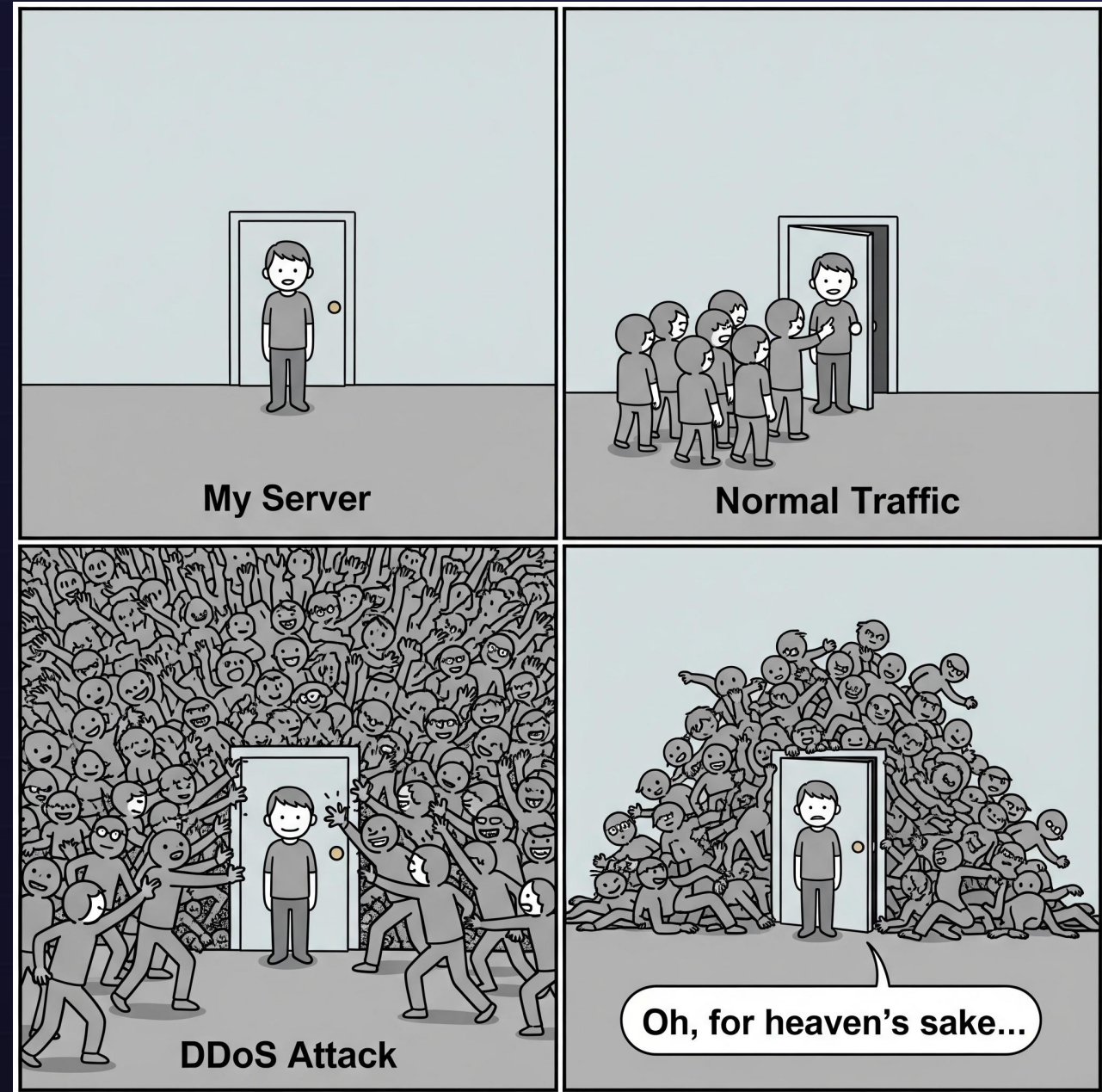
Adaptive Machine Learning Based Real-Time Detection of DDoS Attacks

PRESENTED BY

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| Context: Protecting Modern Networks



Real-Time Responsiveness

Systems must detect and respond to attacks instantly, adapting to dynamic network environments like SDN and cloud infrastructure.



Proactive Mitigation

Beyond simple detection, the primary goal is developing systems capable of automatically initiating countermeasures to block ongoing attacks.



Limited Availability of Real-World Datasets

Training effective models requires authentic, diverse DDoS datasets, which are increasingly difficult to obtain or simulate accurately.



Evolving Threats

Attackers constantly innovate their strategies, requiring detection systems to be robust and adaptable to novel and sophisticated DDoS methods.



State of the Art

A comparative analysis of recent DDoS detection
methodologies



Comparison of Literature Methodologies

Reference	Environment & Dataset	ML Methods Used	Key Strengths & Limitations
Paper 1	CICDDoS2019 dataset (Offline Analysis)	Random Forest, AdaBoost, XGBoost with PCA	Strength: High accuracy using tree-based ML. Limit: No real-time deployment or deep learning.
Paper 2	Kaggle SDN dataset (104,345 records)	Random Forest, SVM, XGBoost	Strength: High accuracy for SDN traffic. Limit: Tested on a single dataset, no real implementation.
Paper 3	Live network monitoring (Wireshark)	Machine learning with automated blocking	Strength: Practical real-time detection and response. Limit: Limited scalability, uses traditional ML only.
Paper 4	SDN Testbed (Ryu controller + Open vSwitch)	Hybrid Model (Random Forest + XGBoost)	Strength: ~99.36% accuracy utilizing SDN flow features. Limit: Tested exclusively in controlled lab environments.

Gap Analysis

Identifying critical limitations in current network defense
literature

Identified Limitations



Lack of Real-time Deployment

Most evaluation is strictly confined to offline datasets or simulated environments. Models lack practical validation in live production environments and real network scenarios.



Limited Dataset Diversity

Studies often heavily rely on single or outdated datasets. These narrow benchmarks fail to account for the highly diverse, encrypted, and complex modern web traffic.



Passive Detection Without Response

A vast majority of systems remain strictly passive. While they are capable of detecting network anomalies, they lack the closed-loop automation required to actively block threats.



Limited Adaptability to Evolving Attacks

Existing setups show weak defensive resilience against zero-day exploits, dynamic multi-vector DDoS strategies, and sophisticated, polymorphic network threats.



Proposed Plan

Strategic directions to overcome existing research gaps



Strategic Focus

Targeted Research Gap	Implementation Strategy	Strategic Justification
Detection with Automatic Mitigation	Add a closed-loop system that automatically blocks the attacker IP or drops traffic immediately upon detection.	Leverages existing, highly accurate ML detection models and simply extends them with an automated mitigation mechanism.
Use of Multiple Datasets	Train, validate, and test the models using multiple distinct DDoS datasets rather than a single source.	Improves model accuracy, robustness, and real-world generalization without requiring complex architectural changes.
Use of Deep Learning Models	Apply neural network architectures such as CNNs, LSTMs, or transformers for sequence and pattern recognition.	Enhances detection of complex attack patterns.

Our Key Contributions (Expected)



Adaptive ML Pipeline

01

A dynamic machine learning framework built to detect and adapt to evolving network threats in real time.



Cross-Dataset Eval

02

Validation across multiple DDoS datasets.



Real-Time Mitigation

03

Closed-loop, automated architecture engineered to instantly block threat sources and drop malicious packets upon detection.



DL Performance Comparison

04

An in-depth, structured performance comparison highlighting the trade-offs in speed and pattern recognition among CNNs, LSTMs, and transformers.



Deployment-Ready Architecture

05

Practical system design optimized for deployment-ready architecture suitable for real-time environments .

Thank You

Our servers survived the presentation. Any questions regarding the research plan?

